

MANIFESTINGO

COFFEE

Customer Relationship Management System

OOP Project • AI-Powered Predictions • 250+ Customers

The Problem

Coffee shops lose customers because they don't know them.

- ✗ No record of what each customer prefers
- ✗ No way to predict who will return and buy again
- ✗ No data to decide which product to promote
- ✗ Customer habits (like addiction level) are ignored

Our Solution



Track every customer's coffee preferences & habits



Store full purchase and visit history



AI engine predicts what each customer will buy next



Probability breakdown: coffee type, cake, accessory



Add new customers via a clean 3-column form

System Architecture

4 Python classes • JSON persistence • AI prediction engine

Coffee

type, name, size,
additions, price, date

Visit

date, type, items,
duration, note

Customer

full profile +
purchases[] + visits[]

CoffeeShop

manages all
customers

JSON — data saved to disk, persists across restarts

AI Engine — scores each customer → probability output

How It Works?

1

Data Collection

Customer profile is created with personal info, coffee habits and addiction level. Every purchase and visit is recorded.

2

Data Storage

All data is saved to data.json automatically. When the program restarts, it loads from the file — nothing is lost.

3

AI Scoring

`predict_customer()` analyses addiction level, visit frequency, purchase history and income to generate a likelihood score (0–100).

4

Prediction Output

The system outputs a probability distribution: which coffee (e.g. Americano 66%), cake likelihood (e.g. 54%), and accessory chance.

Customers Page

MANIFESTINGO COFFEE	Customers					AI Prediction				
	Total Customers 250 Total Purchases 3077 Total Visits 4234 Addicts 65 High Priority 171					Add Customer Full Name Age State Job Addiction Cups/Day Fav Coffee Purchases Visits				
	Customers					AI Prediction				
Robert Mitchell Matthew Robinson Helen Evans Patricia Davis Ryan Harris Kathleen Davis Nicole Taylor Linda Jackson Nicholas Jones Samuel Nelson Patricia Lee Elizabeth Anderson Jennifer Jackson Pamela Hill Stephen Lewis Andrew Scott Eric Lee Larry Thomas William Miller Mark Parker Samantha Ramirez Ashley Allen Jason Taylor Angela Smith Carol Evans										

AI Prediction Page

Coffee probability bars • Cake & accessory likelihood • 20 per page

Purchase Probability & Product Recommendation

Based on purchase history, addiction level, visit behavior and lifestyle.

Showing 1–20 of 250 customers | Page 1

Jennifer Jackson

● 100% chance of buying coffee

☕ What will they order?

Americano		%66.1
Irish Coffee		%8.6
Cappuccino		%5.9
Mocha		%3.2
Other Drinks		%16.2

🍰 Cake alongside?

Any cake %29

🛒 Accessory purchase?

Accessory %3

Analysis: coffee addict — drinks 4 cups/day • very frequent visitor • recently browsed — likely deciding • loyal buyer — 30 purchases • high income — premium products suitable

Add Customer Form

3-column layout • Dropdowns for all selections • Coffee profile included

New Customer Registration

Personal Info

First Name

Last Name

Gender

Age

Marital Status

Has Kids

Number of Kids

State

Contact & Work

Phone

Email

Home Address

Work Address

Job Title

Monthly Salary (\$)

Has a Car

Coffee Profile

Addiction Level

Cups Per Day

Preferred Time

Favorite Coffee

Favorite Cake

Accessory Interest

Social Activities

☐ Reading

☐ Art

☐ Photography

☐ Music

☐ Travel

☐ Yoga

☐ Fitness

☐ Cycling

☐ Hiking

☐ Gaming

☐ Cooking

☐ Golf

☐ Sailing

☐ Cinema

☐ Writing

Save Customer

Customer Detail Page

Full profile • Coffee habits • AI recommendation • Full history

Matthew Robinson

← Back

Personal Information

Gender	Male
Age	42
Marital Status	Married
Children	Yes (3)
State	New Jersey
Phone	+1 471 242 5040
Email	matthew.robinson96@email.com
Job	Marketing Manager
Monthly Salary	\$9,505
Has Car	No
Activities	Sailing

Coffee Profile

Addiction Level	Regular
Cups Per Day	2
Preferred Time	Morning
Favorite Coffee	Ristretto
Favorite Cake	Cheesecake
Accessories	No

AI Prediction

71% chance of buying coffee

Top recommendation: Ristretto (75.8%)

Purchase History

🕒	2025-11-12	—	Ristretto (Small)	—	no additions	—	\$4.62
🕒	2025-06-04	—	Espresso (Large)	—	Vanilla Syrup, No Sugar	—	\$7.31
🕒	2024-08-17	—	Ristretto (Small)	—	Oat Milk, No Sugar	—	\$7.25
🕒	2024-07-19	—	Flat White (Large)	—	Oat Milk	—	\$7.28
🕒	2023-03-09	—	Ristretto (Small)	—	Soy Milk, Vanilla Syrup	—	\$5.49
🕒	2023-02-28	—	Ristretto (Large)	—	Soy Milk, Almond Milk	—	\$5.37
📦	2022-01-04	—	Coffee Bean Bag 1kg (-)	—	no additions	—	\$22.30
🕒	2020-01-19	—	Ristretto (Small)	—	no additions	—	\$7.04
🕒	2018-10-03	—	Ristretto (Small)	—	Almond Milk, Hazelnut Syrup	—	\$6.80
🕒	2018-05-22	—	Ristretto (Large)	—	Hazelnut Syrup, Whipped Cream	—	\$4.71
📦	2017-09-25	—	Ceramic Mug (-)	—	no additions	—	\$37.75
🕒	2016-03-21	—	Ristretto (Large)	—	no additions	—	\$5.42
📦	2015-04-03	—	Designer Mug Set (-)	—	no additions	—	\$15.05

Visit History

- 2025-12-11 — purchase (55 min)
 - 📍 First visit, loved the vibe
- 2025-11-10 — purchase (101 min)
 - 📍 Stayed to work remotely
- 2023-01-24 — purchase (11 min)
 - 📍 Tried a new drink
- 2022-12-02 — just looking (87 min)
 - 📍 Stayed to work remotely
- 2022-09-28 — purchase (12 min)
 - 📍 Checked seasonal menu
- 2022-04-28 — cake (61 min)

OOP Concepts Applied

Encapsulation

Each class manages its own data. Customer holds `purchases[]` and `visits[]` internally.

Abstraction

`predict_customer()` hides complex scoring logic. The UI just calls it and gets a clean result.

Composition

Customer contains Coffee and Visit objects. CoffeeShop contains Customer objects.

Methods

`add_purchase()`, `add_visit()`, `find_customer()`, `remove_customer()` — meaningful behaviors.

Thank You

Manifestingo Coffee CRM

Project Team

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Ege Altınışık

Arda Hacet

Built with Python • OOP • Tkinter • AI Prediction Engine